

Title: Toxic Comment Classifier

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Literature Survey:

Serial No	Year	Publication By	Title	Dataset	Balance	Classes	Algorithm	Metrics	Advantages	Disadvantages
1	2018	IEEE	Deep Learning for Toxic Comment Classification	Kaggle Toxic Comments	Unbalanced	2	CNN	F1 Score, Accuracy	High accuracy with CNNs	Requires large training data
2	2019	Springer	A Hybrid Approach for Toxic Comment Detection	Twitter dataset	Unbalanced	3	LSTM + CNN	F1 Score, Precision, Recall	Combines sequential and spatial features	Complexity in model design
3	2020	IEEE	Multilingual Toxic Comment Classification	Multilingual dataset	Unbalanced	2	Transformer	Accuracy, AUC	Addresses multilingual challenges	Limited data for less common languages
4	2020	Springer	An Ensemble Model for Online Toxicity Detection	Reddit comments	Unbalanced	2	Ensemble Learning	F1 Score, ROC Curve	Robust against noise in data	May overfit on small datasets
5	2021	IEEE	Fine-Tuning BERT for Toxic Comment Classification	Jigsaw dataset	Unbalanced	2	BERT	F1 Score, Precision	State-of-the-art performance	Computationally expensive
6	2021	Springer	Toxic Comment Detection with Contextual Embeddings	Various online platforms	Balanced	3	Contextualized Word Embeddings	F1 Score, Recall	Effective context capture	May not generalize to all domains

7	2022	IEEE	Detecting Toxic Comments Using Deep Learning Techniques	Kaggle dataset	Unbalanced	2	GRU	Accuracy, Precision	Good for sequential data	Sensitive to hyperparameters
8	2022	Springer	Toxicity Detection in Online Reviews	Yelp dataset	Unbalanced	2	Bi-LSTM	F1 Score, Accuracy	Suitable for varied review formats	Limited to reviews
9	2022	IEEE	Real-Time Toxic Comment Detection in Social Media	Live social media feeds	Unbalanced	2	Stream Processing Framework	Latency, F1 Score	Real-time detection capabilities	High processing demands
10	2023	Springer	Addressing Bias in Toxic Comment Classification	Twitter & Reddit	Unbalanced	2	SVM	F1 Score, Confusion Matrix	Addresses fairness issues	Focus on bias limits application scope
11	2023	IEEE	A Comparative Analysis of Toxicity Detection Techniques	Various datasets	Unbalanced	2	Various	F1 Score, ROC Curve	Comprehensive evaluation of methods	Varying results based on dataset quality
12	2023	Springer	Explainable AI for Toxic Comment Detection	Annotated datasets	Balanced	2	Explainable Models	Explainability Metrics, F1 Score	Increases transparency	Complexity in explaining results
13	2023	IEEE	Cross-Domain Toxicity Detection	Multi-domain dataset	Unbalanced	3	Transfer Learning	F1 Score, Accuracy	Effective across different domains	Transfer learning can introduce bias
14	2023	Springer	A Neural Approach to Detect Sarcasm in	Sarcastic dataset	Unbalanced	2	Deep Learning Models	F1 Score, Precision	Innovative handling of sarcasm	Sarcasm detection can be inconsistent

			Toxic Comments							
15	2023	IEEE	Fairness in AI: Addressing Toxicity in Online Comments	Multi-source comments	Balanced	2	Ethical AI Framework	Fairness Metrics, Accuracy	Focus on ethical AI practices	Ethical considerations may limit application
16	2023	Springer	Temporal Analysis of Toxicity in Online Platforms	Longitudinal dataset	Unbalanced	3	Time Series Analysis	Temporal Accuracy, F1 Score	Captures dynamics of online behavior	Data collection can be challenging
17	2023	IEEE	Visualizing Toxic Comments with NLP	Annotated visual dataset	Balanced	2	NLP + Visualization	Visualization Metrics, F1 Score	Engaging understanding of toxicity	Requires user training for interpretation
18	2023	Springer	Ethical Considerations in Toxic Comment Detection	Various sources	Balanced	2	Ethical AI Approaches	Fairness Metrics, Accuracy	Addresses ethical concerns	Ethical considerations may limit generalizability
19	2023	IEEE	Predicting Toxicity in Online Discussions	Forum dataset	Unbalanced	2	SVM + Neural Networks	F1 Score, Confusion Matrix	Good accuracy with hybrid approach	Sensitive to dataset imbalance
20	2023	Springer	Exploring Contextual Factors in Toxic Comment Detection	Social media datasets	Unbalanced	3	Contextualized Embeddings	F1 Score, Accuracy	Effective for diverse comment types	Requires substantial contextual data